ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PRACTICE EXAM 1

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 8 year: numbers and finance (HCF/LCM, simple interest, hire purchase), integers and exponent laws, fractions and decimals, patterns, algebraic expressions and equations, geometry of straight lines and triangles, the Theorem of Pythagoras, circles, surface area and volume, transformations, data handling and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone will not get full marks.
4. A non-programmable calculator may be used, unless a question says otherwise.
5. Round answers to TWO decimal places where necessary. Diagrams are NOT drawn to scale.
6. Give reasons for your statements in geometry questions.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 $(-3)^3$ = …

A 9 B −9 C −27 D 27

1.2 $3{,}2 \times 10^4$ written as an ordinary number is …

A 32 000 B 320 C 3 200 D 320 000

1.3 $\sqrt{(9/16)}$ = …

A $9/4$ B $3/16$ C $81/256$ D $3/4$

1.4 $2x^2 \times 3x^3$ = …

A $5x^5$ B $6x^5$ C $6x^6$ D $5x^6$

1.5 If $3x - 4 = 11$, then x = …

A $7/3$ B 15 C 21 D 5

1.6 A right-angled triangle has shorter sides of 6 cm and 8 cm. The length of the hypotenuse is …

A 14 cm B 10 cm C 48 cm D 100 cm

1.7 Adjacent angles on a straight line add up to …

A 90° B 270° C 180° D 360°

1.8 The circumference of a circle with a radius of 7 cm is … (use $\pi = 22/7$)

A 44 cm B 22 cm C 154 cm D 49 cm

1.9 For any number $a \neq 0$, a^0 = …

A 0 B 1 C a D 10

1.10 The range of 12 ; 7 ; 19 ; 4 ; 10 is …

A 10 B 12 C 4 D 15

[10]

QUESTION 2: NUMBERS AND FINANCE

2.1 Write 240 and 360 as products of prime factors. Hence find the HCF and the LCM of 240 and 360. (4)

2.2 Thandeka invests R8 000 at 9% simple interest per year for 4 years. Calculate the interest and the total amount after 4 years. (3)

2.3 A fridge costs R6 500 cash. On hire purchase, the buyer pays a deposit of 10% of the cash price and then 18 monthly instalments of R390.

2.3.1 Calculate the deposit. (1)

2.3.2 Calculate the total hire-purchase price. (2)

2.3.3 How much more does the buyer pay by buying on hire purchase? (1)

[11]

QUESTION 3: INTEGERS AND EXPONENTS

3.1 Calculate WITHOUT a calculator: (a) $-8 \times (-5) + (-12) \div 4$ (b) $(-2)^4 - (-3)^2$ (c) $\sqrt[3]{(-64)} + \sqrt{49}$ (6)

3.2 Simplify: (a) $(2a^3)^2$ (b) $15x^6 \div 3x^2$ (c) $(5x)^0 + 3^0$ (4)

[10]

QUESTION 4: FRACTIONS AND DECIMALS

4.1 Calculate WITHOUT a calculator. Give answers in the simplest form.

4.1.1 $2\frac{1}{3} + 1\frac{3}{4}$ (3)

4.1.2 $\frac{3}{5} \div \frac{9}{10}$ (2)

4.1.3 $(\frac{2}{3})^2 - \frac{1}{9}$ (2)

4.2 Calculate WITHOUT a calculator: (a) 1,25 × 0,24 (b) 7,2 ÷ 0,09 (3)

[10]

QUESTION 5: PATTERNS, ALGEBRAIC EXPRESSIONS AND EQUATIONS

5.1 Consider the pattern: 5 ; 8 ; 11 ; 14 ; …

5.1.1 Determine the general term (T_n) of the pattern. (2)

5.1.2 Calculate T_{25}. (1)

5.1.3 Which term of the pattern is equal to 152? (2)

5.2 Copy and complete the table for $y = -2x + 1$.

x	−2	−1	0	2	*p*
y	(a)	3	1	(b)	−9

Find the values of (a), (b) and *p*. (3)

5.3 Simplify: (a) $3x^2 - 5x + 2x^2 + 7x$ (b) $-2x(3x - 4)$ (c) $(12a^3 - 8a^2) \div 4a$ (6)

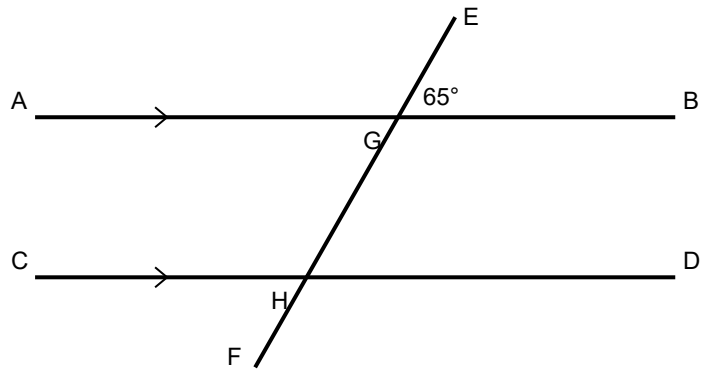
5.4 Solve for x: (a) $5x + 7 = 2x - 8$ (b) $2(x - 3) = 14$ (c) $3^x = 81$ (5)

5.5 Write an equation and solve it: three times a number, decreased by 7, is equal to the number increased by 9. (1)

[20]

QUESTION 6: GEOMETRY OF STRAIGHT LINES AND TRIANGLES

6.1 In the diagram, AB ∥ CD. The line EF cuts AB at G and CD at H. ∠EGB = 65°.



Calculate, giving reasons: (a) ∠GHD (b) ∠BGH (c) ∠GHC (6)

6.2 In ΔPQR, ∠P = $2x$, ∠Q = $3x$ and ∠R = $4x$. Calculate x and the size of each angle. (3)

6.3 In ΔKLM, side LM is extended to N. ∠K = 48° and ∠L = 67°. Calculate ∠KMN. Give a reason. (2)

6.4 Write down the condition for congruency (e.g. SSS) when two triangles have: (a) all three pairs of sides equal; (b) two pairs of sides and the included angle equal. (2)

[13]

QUESTION 7: THEOREM OF PYTHAGORAS AND MEASUREMENT

7.1 A 5 m ladder leans against a vertical wall. The foot of the ladder is 1,4 m from the wall. Calculate how high up the wall the ladder reaches. (3)

7.2 A circle has a radius of 14 cm. Use $\pi = {}^{22}\!/_{7}$ to calculate its (a) area and (b) circumference. (4)

7.3 A triangular prism has a right-angled triangle as its cross-section, with sides of 3 cm, 4 cm and 5 cm. The prism is 10 cm long. Calculate its (a) volume and (b) total surface area. (5)

[12]

QUESTION 8: TRANSFORMATIONS

8.1 Point A is at (3 ; −2). Give the coordinates of the image of A after each transformation:

8.1.1 A reflection in the x-axis (1)

8.1.2 A reflection in the y-axis (1)

8.1.3 A translation of 4 units to the left and 3 units up (2)

8.1.4 A reflection in the line $y = x$ (1)

8.1.5 An enlargement by a scale factor of 2 with the origin as centre (1)

[6]

QUESTION 9: DATA HANDLING AND PROBABILITY

9.1 The marks (out of 100) of 10 learners are: 45 ; 62 ; 58 ; 71 ; 62 ; 80 ; 39 ; 62 ; 55 ; 66

9.1.1 Calculate the mean. (2)

9.1.2 Determine the median. (2)

9.1.3 Write down the mode. (1)

9.1.4 Calculate the range. (1)

9.2 A dice is rolled 60 times and a 4 comes up 12 times. (a) What is the relative frequency of rolling a 4? (b) What is the theoretical probability of rolling a 4? (2)

[8]

TOTAL: 100

END OF QUESTION PAPER